SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 10

Code of Conduct

I Ankala Lakshmi (192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Lakshmi*

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1.Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

A multinational retail organization usually collects customer information from many disconnected sources such as websites, mobile applications, physical stores, loyalty programs, social media, and online payments. The main problem is that these systems may store data in different formats, making it difficult to understand the complete customer journey. A suitable solution is to build a unified big data architecture that combines all customer data into a common platform.

The architecture can begin with an ingestion layer using technologies such as Apache Kafka to collect real-time events from websites, mobile apps, point-of-sale systems, and other sources. Both structured and unstructured data can then be stored in a data lake using technologies such as Hadoop HDFS or cloud object storage. A processing layer using Apache Spark can clean, transform, and combine the data. A customer master-data system can assign a unique customer identity so that the same customer is recognized across online and offline channels.

For real-time analytics, Kafka and Spark Streaming can process activities such as product searches, purchases, cart additions, and store visits immediately. Processed data can be stored in analytical databases or data warehouses for reporting and historical analysis. Machine learning models can then analyze customer behavior and generate personalized product recommendations.

Thus, the proposed architecture provides a single customer view, reduces data fragmentation, supports real-time analytics, and enables the organization to deliver personalized offers and recommendations across different channels.

2.Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Scientific research produces enormous amounts of data from satellites, laboratories, medical experiments, simulations, sensors, and high-performance instruments. Processing this data using a centralized system can become slow because a single machine has limited CPU, memory, and storage capacity.

A distributed computing model solves this problem by dividing a large dataset or computation into smaller tasks and distributing them among multiple computers. Frameworks such as Apache Hadoop and Apache Spark can divide the workload and process different portions of data simultaneously.

For example, suppose a research dataset contains one petabyte of experimental data. Instead of one machine processing the entire dataset sequentially, the data can be divided among hundreds of nodes. Each node processes its assigned portion in parallel, and the results are combined at the end.

Parallel processing improves computational efficiency because multiple processors work at the same time. It also improves scalability, since additional machines can be added when the workload increases. Distributed systems can also provide fault tolerance by replicating data across nodes.Therefore, distributed computing is more suitable than centralized processing for large-scale scientific research because it reduces processing time, supports massive datasets, and allows computing resources to grow according to the research requirements.

3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Digital payment platforms process thousands or millions of transactions every second. Traditional fraud detection systems that analyze transactions only after they are completed may detect fraudulent activity too late. The hidden challenge is therefore to identify suspicious transactions while they are happening.A suitable fraud detection framework should contain a real-time data ingestion layer, streaming processing system, feature-generation layer, predictive model, and alert mechanism.

Transaction information such as amount, location, device, merchant, transaction frequency, and time can be continuously collected using Apache Kafka. A streaming framework such as Spark Streaming or Apache Flink can process these events in real time. The system can calculate features such as unusual transaction frequency, sudden changes in location, repeated failed transactions, or abnormal spending patterns.

Machine learning models such as Random Forest, Logistic Regression, Gradient Boosting, or anomaly-detection algorithms can assign a fraud probability to every transaction. Transactions with high-risk scores can be blocked or sent for additional verification.Streaming processing reduces the detection delay, while predictive models identify complex patterns that simple rule-based systems may miss. Combining both approaches improves fraud detection accuracy and allows suspicious transactions to be handled before significant financial damage occurs.

4 .Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Multimedia data such as videos, audio recordings, and images creates several challenges because it has a much larger storage requirement than ordinary structured data. It also has different formats, high bandwidth requirements, and complex processing requirements.

A traditional relational database is generally not suitable for storing huge volumes of raw multimedia files. Instead, a distributed file system or cloud object storage should be used. Technologies such as Hadoop HDFS, Amazon S3, Azure Blob Storage, or Google Cloud Storage can store large multimedia objects reliably.

Metadata such as file name, user information, timestamp, format, location, and content description can be stored separately in databases such as MongoDB or relational databases. This allows applications to search and manage multimedia files efficiently without storing the entire media content inside a relational table.

For processing, distributed frameworks such as Apache Spark can be used, while specialized tools can perform image recognition, video analysis, speech processing, and compression. Data can also be compressed and stored in suitable formats to reduce storage and transmission costs.

Therefore, an effective multimedia big data environment should separate large raw media storage from metadata management, use distributed or cloud storage, and apply parallel processing for large-scale analysis.

5.Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

A cloud-based big data platform may contain critical business information, so a hardware failure, cyberattack, software problem, or cloud-region outage should not stop the entire system. The main objective of disaster recovery is to restore services quickly while minimizing data loss.The platform should first use redundant infrastructure so that critical services do not depend on a single server. Data should be replicated across multiple availability zones or geographical regions. Technologies such as cloud object storage replication, database replication, and distributed storage replication can maintain multiple copies of important data.

Regular backups should also be performed and stored separately from the primary system. A disaster recovery plan should define Recovery Point Objective (RPO), which specifies the acceptable amount of data loss, and Recovery Time Objective (RTO), which specifies how quickly the service must be restored.

Automated monitoring can detect failures and trigger automatic failover to a backup server or region. Load balancers and health checks can redirect users to healthy services.

Thus, redundancy prevents a single point of failure, replication protects data, and automated failover reduces downtime. Together, these mechanisms provide business continuity and operational resilience.

6.Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Data quality is extremely important in large-scale analytics because analytical results are only as reliable as the data used to produce them. Common problems include **missing values, duplicate records, inconsistent formats, incorrect values, outdated information, and conflicting records**.

For example, if the same customer is stored under slightly different names in different systems, the organization may incorrectly treat one person as two customers. Similarly, missing or incorrect transaction values can produce inaccurate business reports and machine learning predictions.

To solve these problems, the organization should implement a strong **data governance framework**. Data validation rules should be applied during data ingestion. ETL or data-processing pipelines can identify duplicates, standardize formats, handle missing values, and detect invalid records.

The organization should also establish **data ownership and stewardship**, where responsible teams define data standards and monitor quality. Data catalogs and metadata management can help users understand the origin and meaning of datasets. Regular data-quality audits should measure completeness, accuracy, consistency, and timeliness.

Therefore, data governance is not simply a technical activity. It requires proper processes, responsibilities, standards, and monitoring to ensure that analytical decisions are based on trustworthy data.

7.Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

An online streaming platform such as a video or music service generates enormous amounts of user behavior data. This includes viewing history, search activity, watch duration, likes, skips, ratings, and browsing behavior. The challenge is to analyze this information at scale and recommend content that users are actually interested in.

A scalable recommendation system can use Kafka to collect user events in real time. These events can be stored in a data lake and processed using Apache Spark. The system can use collaborative filtering to identify users with similar behavior and recommend content based on their preferences.

For example, if thousands of users who watched Movie A also watched Movie B, the system can identify this relationship and recommend Movie B to users who have watched Movie A. Content-based filtering can also consider attributes such as genre, actors, language, and keywords.

Machine learning models can combine user behavior and content information to produce personalized recommendations. Distributed processing allows large numbers of users and content items to be processed simultaneously.

Therefore, big data technologies make the recommendation system scalable, while user behavior analytics improves personalization. Better recommendations increase user satisfaction, watch time, and ultimately customer retention.

8.Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Cloud computing has become an important part of modern big data systems because organizations can obtain large amounts of computing and storage resources without purchasing and maintaining physical infrastructure. One major advantage is scalability. Cloud platforms allow organizations to increase computing resources when data volumes or workloads increase and reduce them when demand falls. This is particularly useful for big data applications where workloads can change significantly.

Cloud computing also provides flexibility because organizations can use services for storage, databases, data processing, machine learning, and analytics according to their requirements. The pay-as-you-go model can reduce the need for large initial infrastructure investments.

However, cloud computing also has limitations. Continuous usage can become expensive if resources are not monitored properly. Data transfer costs, vendor dependency, network failures, and security concerns are other challenges. Organizations must also carefully manage access permissions and encryption because sensitive data is being stored on third-party infrastructure.

Therefore, cloud computing provides significant scalability and flexibility for big data, but organizations need proper cost management, security controls, monitoring, and governance to gain its benefits without creating new risks.

9.Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Transportation and logistics companies generate continuous data from GPS devices, vehicles, traffic systems, warehouses, fuel sensors, delivery applications, and customer orders. The major challenge is converting this continuously generated data into immediate operational decisions. A real-time analytics framework can use IoT sensors and GPS devices as data sources. Data can be collected through Apache Kafka and processed using Spark Streaming or Apache Flink. The system can combine vehicle location, traffic conditions, weather, fuel consumption, delivery schedules, and historical route information .Analytics and machine learning models can identify the best route based on traffic conditions, delivery priority, distance, and fuel consumption. If an accident or traffic congestion occurs, the system can immediately recalculate the route and provide a new recommendation to the driver.Historical data can also be analyzed to identify frequently congested routes, fuel-wasting behavior, and delivery delays. Predictive analytics can help organizations estimate delivery times and perform preventive vehicle maintenance.As a result, real-time big data analytics can reduce travel time, fuel consumption, vehicle idle time, and delivery delays while improving operational decision-making.

10.Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Big data provides organizations with valuable insights, but collecting and analyzing large amounts of personal information creates serious ethical and legal concerns. Organizations may collect information about users' behavior, location, preferences, purchases, and online activities without users fully understanding how the information will be used.

One major concern is privacy. Personal information should not be collected or processed beyond what is necessary for a legitimate purpose. Another issue is security because a large centralized dataset becomes an attractive target for attackers. Poor-quality or biased data can also lead to unfair decisions, particularly when machine learning is used for recruitment, lending, healthcare, or insurance.

Organizations should therefore establish clear data governance and privacy policies. Data collection should follow principles such as transparency, purpose limitation, data minimization, and appropriate consent. Sensitive data should be protected through encryption, access control, authentication, and secure storage.

Organizations should also conduct regular security and compliance audits and maintain clear data-retention and deletion policies. Machine learning systems should be tested for bias and unfair outcomes, while users should be informed about important automated decisions where appropriate.

Compliance with applicable privacy and data-protection laws is essential, but legal compliance alone is not enough. Organizations must build a culture of responsible data usage, transparency, security, and accountability. This helps protect users and maintain long-term trust in the organization.